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Compact electric drive units and vehicles

Abstract

Compact electric drive units and vehicles with such drive units are provided. An exemplary electric drive unit includes an electric motor configured to provide an electric motor torque; an input differential configured to split the electric motor torque into a first path and a second path; a final drive gear on each path; a pinion gear on each path; and a rear transfer shaft assembly and a front transfer shaft assembly on each path. Further, each pinion gear is configured to transfer torque to a respective rear transfer shaft assembly and a respective front transfer shaft assembly, and the respective rear transfer and front transfer shaft assemblies are configured to transfer torque to the respective final drive gear.

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Background/Summary

INTRODUCTION

(1) The disclosure relates to a multiple-axis torque sharing electric drive unit that employs a single electric motor to drive a pair of opposite side wheels in a vehicle.

(2) Modern motor vehicles are typically configured as either two-wheel-drive or all-wheel-drive. Either type of a vehicle may employ a conventional powertrain, where a single engine is used to propel the vehicle, an electric powertrain, where an electric motor is used to propel the vehicle, or a hybrid powertrain, where two or more distinct power sources, such as an internal combustion engine and an electric motor, are used to accomplish the same task.

(3) An all-wheel-drive hybrid vehicle may be configured as an axle-split vehicle. In such a vehicle, independent power-sources, such as an internal combustion engine and an electric motor, are set up to independently power individual vehicle axles that are operatively connected to the respective

power-sources, thus generating on-demand all-wheel-drive propulsion. In such an axle-split hybrid vehicle employing an engine and an electric motor, the electric motor may be capable of propelling the vehicle via the respective axle while the engine is shut off.

(4) Each powered axle typically includes a final drive assembly with a differential that allows opposite side, i.e., left (driver) and right (passenger) side, driven wheels to rotate at different speeds when the vehicle negotiates a turn. Specifically, the differential permits the driven wheel that is traveling around the outside of the turning curve to roll farther and faster than the driven wheel traveling around the inside of the turning curve, while approximately equal torque is applied to each of the driven wheels. An increase in the speed of one driven wheel is balanced by a decrease in the speed of the other driven wheel, while the average speed of the two driven wheels equals the input rotational speed of the drive shaft connecting the power-source to the differential.

(5) The need for new and improved drive units continues with the design of lighter and more compact vehicles, particularly for vehicles driven by electric motors. Accordingly, it is desirable to provide a cost-effective, compact drive unit. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing introduction.

SUMMARY

(6) Compact electric drive units and vehicles with such drive units are provided. An exemplary electric drive unit includes an electric motor configured to provide an electric motor torque; an input differential configured to split the electric motor torque into a first path and a second path; a first final drive gear on the first path and a second final drive gear on the second path; a first pinion gear on the first path and a second pinion gear on the second path; a first rear transfer shaft assembly and a first front transfer shaft assembly on the first path; and a second rear transfer shaft assembly and a second front transfer shaft assembly on the second path; wherein the first pinion gear is configured to transfer torque to the first rear transfer shaft assembly and to the first front transfer shaft assembly; wherein the first rear transfer shaft assembly and the first front transfer shaft assembly are configured to transfer torque to the first final drive gear; wherein the second pinion gear is configured to transfer torque to the second rear transfer shaft assembly and the second front transfer shaft assembly; and wherein the second rear transfer shaft assembly and the second front transfer shaft assembly are configured to transfer torque to the second final drive gear.

(7) An exemplary electric drive unit includes an electric motor configured to provide an electric motor torque; an input differential; a pinion gear, wherein the input differential is configured to selectively transmit torque to the pinion gear; a rear transfer shaft assembly and a front transfer shaft assembly, wherein the pinion gear is configured to transfer torque to the rear transfer shaft assembly and to the front transfer shaft assembly; a final drive gear, wherein the rear transfer shaft assembly and the front transfer shaft assembly are configured to transfer torque to the final drive gear; and a locking device configured to selectively connect and disconnect the input differential from the pinion gear.

(8) An exemplary vehicle includes an electric motor configured to provide an electric motor torque; a first pinion gear on a first path and a second pinion gear on a second path; a first rear transfer shaft assembly on the first path and a second rear transfer shaft assembly on the second path; a first front transfer shaft assembly on the first path and a second front transfer shaft assembly on the second path; a first final drive gear on the first path and a second final drive gear on the second path; and a first wheel on the first path and a second wheel on the second path; wherein the first pinion gear is configured to transfer torque to the first rear transfer shaft assembly and the first front transfer shaft assembly; wherein the first rear transfer shaft assembly and the first front transfer shaft assembly are configured to transfer torque to the first final drive gear; wherein the second pinion gear is configured to transfer torque to the second rear transfer shaft assembly and the second front transfer shaft assembly; and wherein the second rear transfer shaft assembly and

the second front transfer shaft assembly are configured to transfer torque to the second final drive gear.

Description

DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:
- (2) FIG. 1 is a functional block diagram of a vehicle, in accordance with exemplary embodiments;
- (3) FIG. 2 is a perspective view of an electric drive unit for the vehicle of FIG. 1, in accordance with exemplary embodiments;
- (4) FIG. 3 is a front view schematic of the electric drive unit of FIG. 2, in accordance with exemplary embodiments;
- (5) FIG. 4 is an end view schematic of the electric drive unit of FIG. 3;
- (6) FIG. 5 is a focused front view schematic of the locking device of the electric drive unit of FIG. 3 in an unlocked configuration, in accordance with exemplary embodiments;
- (7) FIG. 6 is a focused front view schematic of the locking device of the electric drive unit of FIG. 3 in a locked configuration, in accordance with exemplary embodiments;
- (8) FIG. 7 is a front view schematic of the electric drive unit of FIG. 2, in accordance with exemplary embodiments;
- (9) FIG. 8 is an end view schematic of the electric drive unit of FIG. 7;
- (10) FIG. 9 is a focused front view schematic of the locking device of the electric drive unit of FIG. 7 in an unlocked configuration, in accordance with exemplary embodiments;
- (11) FIG. 10 is a focused front view schematic of the locking device of the electric drive unit of FIG. 7 in a locked configuration, in accordance with exemplary embodiments;
- (12) FIG. 11 is schematic focused on a bearing arrangement utilized on a transfer shaft assembly of FIG. 3, in accordance with exemplary embodiments; and
- (13) FIG. 12 is schematic focused on a bearing arrangement utilized on a final drive assembly of FIG. 3, in accordance with exemplary embodiments.

DETAILED DESCRIPTION

(14) The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding introduction, brief summary or the following detailed description. As used herein, the term module refers to any hardware, software, firmware, electronic control unit or component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

(15) Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by any number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with any number of automated driving systems including cruise control systems, automated driver assistance systems and autonomous driving systems, and that the vehicle system

described herein is merely one example embodiment of the present disclosure.

(16) For the sake of brevity, conventional techniques related to signal processing, data transmission, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

(17) Spatially relative terms, such as “front”, “rear”, “above”, “below”, “behind” “top”, “bottom”, “under”, “over”, “side”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in a figure or figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(18) Various connections, such as between a driving gear and a driven gear, and between a gear and an axial shaft, are described herein. In exemplary embodiments, a connected driving gear and driven gear mesh with one another. Specifically, each gear has cut gear teeth, and the cut gear teeth mate with one another such that the driving gear may directly contact, engage with, and drive the driven gear. Thus, the connection therebetween is provided between the cut gear teeth. In exemplary embodiments, a gear that is coupled or mounted to an axial shaft is directly fixed to the axial shaft or sleeve, such the gear and shaft or sleeve rotate together. In certain embodiments, cut teeth are formed on a shaft to define a gear that is coupled or mount to the shaft.

(19) Referring to FIG. 1, a vehicle **40** is illustrated in functional block diagram form. In various embodiments, the vehicle **40** is an automobile. The vehicle **40** may be any one of a number of different types of automobiles, such as, for example, a sedan, a wagon, a van, a truck, or a sport utility vehicle (SUV), and may be two-wheel drive (2 WD) (i.e., rear-wheel drive or front-wheel drive), four-wheel drive (4 WD) or all-wheel drive (AWD), and/or various other types of vehicles in certain embodiments. In certain embodiments, the vehicle **40** may also include another type of mobile platform.

(20) As depicted in FIG. 1, the vehicle **40** includes a body **62** that is arranged on or integrated with a chassis. The body **62** substantially encloses other components of the vehicle **40**. The vehicle **40** also includes a plurality of wheels **64** and **65**, which may be referred to as left or driver-side wheels **64** and right or passenger-side wheels **65**. The wheels **64** and **65** are each rotationally coupled to the chassis near a respective corner of the body **62** to facilitate movement of the vehicle **40**. In one embodiment, the vehicle **40** includes four wheels **64** and **65**, although this may vary in other embodiments (for example for trucks and certain other vehicles).

(21) A drive system **68** is mounted in the vehicle **40**, and drives the wheels **64** and **65**, for example via axles **66**, **67**. In certain embodiments, the drive system **68** includes a drive unit **100**. In certain exemplary embodiments, the drive unit **100** includes a powerplant **72**, such as an internal combustion engine and/or an electric motor/generator, that is coupled with a transmission system **400**. In exemplary embodiments, the powerplant **72** is an electric motor/generator. In certain embodiments, the drive system **68** may vary, and/or two or more drive systems **68** may be used. By way of example, the vehicle **40** may also incorporate any one of, or combination of, a number of different types of drive unit **100**, such as, for example, a gasoline or diesel fueled combustion engine, a “flex fuel vehicle” (FFV) engine (i.e., using a mixture of gasoline and alcohol), a gaseous compound (e.g., hydrogen and/or natural gas) fueled engine, a combustion/electric motor hybrid engine, and an electric motor.

(22) As depicted in FIG. 1, the vehicle **40** also includes a braking system **78** and a steering system **80** in various embodiments. In exemplary embodiments, the braking system **78** controls braking of

the vehicle **40** using an actuator **82** that may be controlled via inputs provided by a driver, such as through a brake pedal as the actuator **82**, and in certain embodiments, via automatic control by a control system **84**. The braking system **78** includes brakes, such as brake **85**, at any of the number of wheels **64** and **65**. Also in exemplary embodiments, the steering system **80** controls steering of the vehicle **40** via an actuator **86**, such as with inputs from a steering wheel **88** (e.g., in connection with a steering column coupled to the axle **66** and/or the wheels **64** and **65**), that are controlled via inputs provided by a driver, and in certain embodiments via automatic control via the control system **84**.

(23) In the embodiment depicted in FIG. **1**, the control system **84** is coupled with various systems including the braking system **78** and the steering system **80** of the vehicle **40**. In various embodiments, the control system **84** may also be coupled to one or more other systems and/or components of the vehicle **40** and includes a controller **90**. As illustrated in FIG. **1**, the controller **90**, and the powertrain controller **89** are a part of, or comprise, a computer system **92**. It will be appreciated that the controller **90** may otherwise differ from the example depicted in FIG. **1**. The controller **90** may be configured as any number of controllers and/or microcontrollers in communication with each other.

(24) As illustrated in FIG. **1**, the controller **90** is coupled with various devices and systems of the vehicle **40**, such as the braking system **78** and the steering system **80**. The controller **90** may accept information from various sources, process that information, and provide control commands based thereon to effect outcomes such as operation of the vehicle **40** and its systems, including of the braking system **78**. In the depicted embodiment, the controller **90** includes a processor **94** including a memory device and storage device. The processor **94** performs the computation and control functions of the controller **90**, and may be any type of processor or multiple processors, single integrated circuits such as a microprocessor, or any suitable number of integrated circuit devices and/or circuit boards working in cooperation to accomplish the functions of a processing unit. During operation, the processor **94** may execute one or more programs and may use data, each of which may be stored within the processor **94**, the processor **94** controls the general operation of the controller **90** in executing the processes described herein, such as the processes and methods described in greater detail below.

(25) Still referring to FIG. **1**, the controller **90** is coupled with various actuators including the actuators **82**, **86** and the drive unit **100**. The controller **90** is also coupled with various sensors that sense observable conditions of the vehicle **40**. In this embodiment, the sensing devices include, but are not limited to, a braking sensor such as a pedal position sensor, a steering angle sensor, an acceleration sensor, and a torque request sensor, such as at the accelerator pedal or throttle.

(26) As shown in FIG. **1**, in various embodiments, the controller **90** is coupled to, among other devices, the sensors, the braking system **78**, the steering system **80**, the drive unit **100**, and/or one or more other systems, devices, and/or components of the vehicle **40**.

(27) The controller **90** may provide the drive unit **100** control functions of the vehicle **40**. In embodiments, a powertrain controller **89** may be coupled in the control system **84** and in certain embodiments may be included in the controller **90**. In embodiments, the powertrain control module **91** may reside in, or may comprise, the powertrain controller **89**, which may be a part of the controller **90** or which may be a separate powertrain controller **89** coupled with the controller **90**. The powertrain controller **89** may provide the various functions of controlling the drive unit **100**, such as by providing a torque command to operate the drive unit **100** to propel the vehicle **40**.

(28) In FIG. **2**, a portion of an exemplary drive unit **100** including a portion of power plant **72** and transmission system **400** is illustrated. As shown, a portion of the power plant **72** is an electric motor configured to provide an electric motor torque and includes a rotor **301** that rotates about a rotor axis **303**. The rotor **301** is interconnected with an input differential **600** configured to selectively split the electric motor torque into a first path **401** and a second path **402**. Cross-referencing FIGS. **1** and **2**, the first path **401** extends from the input differential **600** to wheel **64**,

and the second path **402** extends from the input differential **600** to wheel **65**.

(29) In the exemplary embodiment shown in FIG. 2, the input differential **600** is configured to selectively transmit torque from the rotor **301** to two pinion shafts or drive shafts **302** that are each configured to rotate about the rotor axis **303**. Specifically, the drive shafts **302** are coaxial. Further, the drive shafts **302** are each configured to rotate a pinion gear **305**. In exemplary embodiments, each pinion gear **305** is mounted on a respective drive shaft **302**. The pinion gears **305** are configured to rotate about the rotor axis **303**.

(30) As further shown in FIG. 2, each pinion gear **305** is configured to transmit torque to two parallel transfer shaft assemblies **450** and **460**, which may be referred to as rear transfer shaft assembly **450** and front transfer shaft assembly **460**. In the illustrated embodiment, each rear transfer shaft assembly **450** rotates about a rear axis **451** and each front transfer shaft assembly **460** rotates about a front axis **461**.

(31) In FIG. 2, each rear transfer shaft assembly **450** comprises a transfer shaft **550**, a transfer gear **551**, and a final drive pinion **552**. In exemplary embodiments, each transfer gear **551** is formed on or mounted to the respective transfer shaft **550**. In exemplary embodiments, each final drive pinion **552** is formed on or mounted to the respective transfer shaft **550**. As a result, each transfer shaft **550**, transfer gear **551**, and final drive pinion **552** rotates about the rear axis **451**.

(32) In FIG. 2, each front transfer shaft assembly **460** comprises a transfer shaft **560**, a transfer gear **561**, and a final drive pinion **562**. In exemplary embodiments, each transfer gear **561** is formed on or mounted to the respective transfer shaft **560**. In exemplary embodiments, each final drive pinion **562** is formed on or mounted to the respective transfer shaft **560**. As a result, each transfer shaft **560**, transfer gear **561**, and final drive pinion **562** rotates about the front axis **461**.

(33) As further shown in FIG. 2, each pair of parallel transfer shaft assemblies **450** and **460** is configured to transmit torque to a respective final drive assembly **480**. Each final drive assembly **480** includes a final drive gear **482** and a drive shaft or output shaft **485**. Each final drive gear **482** is configured to rotate about a drive axis **481**. Specifically, each final drive gear **482** is coupled or mounted to the drive shaft **485** or output shaft **485** rotatable about drive axis **481**.

(34) Thus, in FIG. 2, first torque path **401** includes an interconnected drive shaft **302**; pinion gear **305**; rear transfer shaft assembly **450**, including transfer shaft **550**, transfer gear **551**, and final drive pinion **552**; final drive gear **482**, and drive shaft **485**.

(35) Further, in FIG. 2, second torque path **402** includes an interconnected drive shaft **302**; pinion gear **305**; rear transfer shaft assembly **450**, including transfer shaft **550**, transfer gear **551**, and final drive pinion **552**; and final drive assembly **480**, including final drive gear **482** and drive shaft **485**.

(36) In each torque path **401** and **402**, the pinion gear **305** is configured to transfer torque to the rear transfer shaft assembly **450** and to the front transfer shaft assembly **460**; and the rear transfer shaft assembly **450** and the front transfer shaft assembly **460** are configured to transfer torque to the final drive assembly **480**. Further, in the torque path **401** and **402**, the final drive assembly **480** is configured to transfer torque to the output shaft or drive shaft **485**.

(37) In certain embodiments, in each torque path **401** and **402**, the pinion gear **305** meshes with the rear transfer shaft assembly **450** and the front transfer shaft assembly **460**; and the rear transfer shaft assembly **450** and the front transfer shaft assembly **460** mesh with the final drive assembly **480**. More specifically, in each torque path **401** and **402**, the pinion gear **305** meshes with the transfer gear **551** of the rear transfer shaft assembly **450**, the pinion gear **305** meshes with the transfer gear **561** of the front transfer shaft assembly **460**, the final drive pinion **552** of the rear transfer shaft assembly **450** meshes with the final drive gear **482** of the final drive assembly **480**, and the final drive pinion **562** of the front transfer shaft assembly **460** meshes with the final drive gear **482** of the final drive assembly **480**.

(38) Each torque path **401** and **402** is provided with a first stage speed reduction from the pinion gear **305** to the rear transfer shaft assembly **450** and to the front transfer shaft assembly **460**. Further, each torque path **401** and **402** is provided with a second stage speed reduction from the

rear transfer shaft assembly **450** and from the front transfer shaft assembly **460** to the final drive assembly **480**.

(39) As shown in FIG. 2, the drive unit **100** includes four axes of rotation: rotor axis **303**, rear axis **451**, front axis **461**, and drive axis **481**, none of which are co-axial with one another.

(40) Referring now to FIGS. 3 and 4, further features of the drive unit **100** are illustrated. FIG. 3 is a front view schematic, such that the view of the rear transfer shaft assemblies **450** is blocked by the front transfer shaft assemblies **460**, and FIG. 4 is an end view schematic showing one rear transfer shaft assembly **450** and one front transfer shaft assembly **460**. FIGS. 3 and 4 illustrate an exemplary drive unit **100** including power plant **72** and transmission system **400**. In exemplary embodiments, the transmission system **400** is configured to transmit power received from the motor **72** to the wheels **64** and **65** (shown in FIG. 1) according to selectable speed ratios.

(41) As shown in FIGS. 3 and 4, the motor **72** includes the rotor **301** and a stator **501**. The stator **501** includes a stator winding **502**. The electric motor **72** selectively passes an electric current through the stator winding **502** to produce a magnetic field causing the rotor **301** to rotate about the rotor axis **303**.

(42) As shown in FIG. 3, the transmission **400** includes a differential gearset or differential **600**. The differential **600** operatively connects the electric motor **72** to the pinion or drive shafts **302**. Thus, the differential **600** may split torque from the motor **72** into two torque paths **401** and **402**.

(43) Each drive shaft **302** extends from the differential **600** to an end **308**. Further, a pinion gear **305** is coupled to or formed on the end **308** of the drive shaft **302**. Each drive shaft **302** and pinion gear **305** is configured to rotate about the rotor axis **303**.

(44) FIG. 3 illustrates that each pinion gear **305** meshes with the transfer gear **561** of the respective front transfer shaft assembly **460** at an interface **590**. FIG. 4 illustrates that the pinion gear **305** meshes with the transfer gear **561** of the respective front transfer shaft assembly **460** at an interface **590** and meshes with the transfer gear **551** of the respective rear transfer shaft assembly **450** at an interface **580**. As shown, each rear transfer shaft assembly **450** is offset from the respective front transfer shaft assembly **460** by 180 degrees. Specifically, the pinion axis **303** is located directly between the interface **580** formed by the rear transfer shaft assembly **450** and the pinion gear **305** and the interface **590** formed by the front transfer shaft assembly **460** and the pinion gear **305**. Separating the mesh interfaces **580** and **590** by 180 degrees about each pinion gear **305** yields a net zero bending load on the pinion shafts **302**. The net zero bending loads on the pinion shafts **302** allow for smaller bearing for support of the shafts **302**. In exemplary embodiments, the smaller bearings yield lower losses and higher efficiency.

(45) Cross-referencing FIGS. 3 and 4, the final drive pinion **562** of each front transfer shaft assembly **460** meshes with the final drive gear **482** of the final drive assembly **480** and the final drive pinion **552** of each rear transfer shaft assembly **450** meshes with the final drive gear **482** of the final drive assembly **480**. Because torque from the pinion gear **305** is shared by the transfer shaft assemblies **450** and **460**, the secondary stage transfer shaft assembly reductions from each transfer shaft assembly **450** and **460** to the final drive assembly **480** also see twenty-five percent of the total motor torque each. This arrangement allows for smaller diameter transfer shaft assembly support bearings which may further minimize losses and increase efficiency.

(46) In the illustrated embodiment of each final drive assembly **480**, each final drive gear **482** is mounted on a respective drive shaft **485**. Further, each final drive gear **482** and drive shaft **485** rotates about the drive axis **481**.

(47) Both front transfer shaft assemblies **460**, including shafts **560**, transfer gears **561**, and final drive pinions **562**, rotate about the front axis **461**. Likewise, both rear transfer shaft assemblies **450**, including shafts **550**, transfer gears **551**, and final drive pinions **552**, rotate about the rear axis **451**. As shown in FIG. 4, the rotor axis **303** is located directly between and equidistant from the front axis **461** and rear axis **451**. Accordingly, rotor axis **303**, front axis **461**, and rear axis **451** define and lie in a single plane. In other words, there is 180 degree alignment between the front axis **461**, rotor

axis **303**, and rear axis **451**. As a result, the transmission **400** is provided with radial load cancelling.

(48) As further shown in FIG. **4**, the plane defined by the front axis **461**, rear axis **451**, and rotor axis **303** is perpendicular to a plane defined by the rotor axis **303** and the drive axis **481**.

(49) In FIG. **3**, the drive unit **100** is provided with a bearing arrangement of ten deep groove ball bearings **307** (10 DGBB) and six cylindrical roller bearings **306** (6 CRB). Specifically, each shaft **302** is axially constrained by an internal bearing press fit wherein the shaft **302** is constrained with two deep groove ball bearings (2 DGBB); each of the two rotor pinion gears **305** is constrained with a spline/snap ring interface on one end and a deep groove ball bearing on the other (2 DGBB); each of the four transfer shaft assemblies **450** and **460** are constrained by on a deep groove ball bearing on one end (4 DGBB) and a cylindrical ball bearing on the other (4 CRB); and each of the two final drive assemblies **480** are constrained by on a deep groove ball bearing (2 DGBB) on one end and a cylindrical ball bearing on the other (2 CRB).

(50) FIGS. **11** and **12** provide focused views of bearing arrangements **1100** utilized on the transfer shaft assemblies **450** and **460** and final drive assemblies **480**, respectively.

(51) FIG. **11** illustrates a bearing arrangement **1100** utilized with both of the rear transfer shaft assemblies **450** and both of the front transfer shaft assemblies **460**. For simplicity of numbering, the transfer shaft assembly is described with numbering associated with rear transfer shaft assembly **450**.

(52) In FIG. **11**, the transfer shaft assembly **450** comprises a transfer shaft **550**, a transfer gear **551**, and a final drive pinion **552**. In exemplary embodiments, the transfer gear **551** is formed on or mounted to the transfer shaft **550**. In exemplary embodiments, the final drive pinion **552** is formed on or mounted to the transfer shaft **550**. As a result, the transfer shaft **550**, transfer gear **551**, and final drive pinion **552** rotate about the axis **451**.

(53) As further shown, the transfer shaft assembly **450** is provided with a bearing arrangement **1100** including deep groove ball bearings **307** adjacent to the transfer gear **551** and cylindrical roller bearings **306** adjacent to the final drive pinion **552**.

(54) FIG. **12** illustrates a bearing arrangement **1100** utilized with both of the final drive assemblies **480**.

(55) In FIG. **12**, the final drive assembly **480** comprises a final drive gear **482** and a drive shaft or output shaft **485**. The final drive gear **482** is configured to rotate about drive axis **481**. Specifically, the final drive gear **482** is coupled or mounted to the drive shaft or output shaft **485** rotatable about drive axis **481**.

(56) As further shown, final drive assembly **480** is provided with a bearing arrangement **1100** including deep groove ball bearings **307** and cylindrical roller bearings **306**. As shown, the final drive gear **482** is located between the deep groove ball bearings **307** and cylindrical roller bearings **306**, with the output shaft **485** extending away from the cylindrical roller bearings **306**.

(57) While FIGS. **11** and **12** illustrate an embodiment of a bearing arrangement **1100**, other suitable arrangements may be utilized. For example, taper roller bearings, needle roller bearings, four-point bearings, or other suitable bearings may be used.

(58) Referring back to FIG. **3**, the drive unit **100** further includes a locking device **800** configured to selectively connect and disconnect the input differential housing **650** from the first pinion shaft **302**. Specifically, in the embodiment of FIG. **3**, the locking device **800** includes an axially actuatable fork **810** configured to move along a fork axis **820** parallel to the rotor axis **303**.

(59) Cross-referencing FIGS. **3**, **5** and **6**, further features of the locking device **800** are illustrated. In FIG. **5**, the locking device is in an unlocked configuration **801**. In FIG. **6**, the locking device **800** is in a locked configuration **802**.

(60) As shown, the exemplary locking device **800** includes a linear actuator **830** and shaft **835** for moving the fork **810** along the fork axis **820**. Further, the locking device **800** includes a sliding collar **840** positioned around the pinion shaft **302**. The collar **840** includes axially-extending teeth

and is configured to move axially, i.e., along axis **303**, but relative rotation between the collar **840** and the shaft **302** is prohibited.

(61) Further, the exemplary locking device **800** includes a biasing element **850**, such as a spring, biased to extend to push the collar **840** to the unlocked configuration **801** shown in FIG. 5. Also, the locking device **800** includes a projection **860** formed on the differential housing **650** and coupled to the differential **600**.

(62) As shown in FIG. 6, locking device **800** is in the locked configuration **802** in which the linear actuator **830** has advanced the fork **810** along axis **820**, causing the collar **840** to advance along axis **303** such that the collar **840** compresses the biasing element **850** and engages the projection **860** of the differential housing **650**. Specifically, the axially-extending teeth of the collar **840** may engage the projection **860**. Due to the engagement between the projection **860** and the collar **840**, the collar **840** and shaft **302** are locked to the projection **860**. Thus, the collar **840** and shaft **302** are locked to and rotate with the differential housing **650**.

(63) When the locking device **800** retracts the fork **810** to the unlocked configuration of FIG. 5, the fork **810** and/or the biasing element **850** move the collar **840** out of connection with the projection **860**. As a result, the collar **840** and shaft **302** are disconnected from the projection **860** and unlocked from the differential housing **650**. Thus, the differential **600** is open and may split torque along both paths **401** and **402**.

(64) Cross-referencing FIGS. 3, 5, and 6, the fork **810**, is located directly between the drive shaft **302** and the stator windings **502** of the stator **501**. Further, the axis **820** of the locking device **800** passes through the rotor **301**, and is not co-axial with the rotor axis **303**.

(65) The drive unit **100** of FIGS. 7-10 includes a similar motor **72** and transmission **400** as the embodiment illustrated in FIGS. 3-6, but is provided with another embodiment of a locking device **900**. FIG. 7 is a front schematic view of the drive unit **100**, similar to FIG. 3. FIG. 8 is an end view schematic of the electric drive unit **100**. FIG. 9 is a focused front view schematic of the locking device **900** in an unlocked configuration **901**. FIG. 10 is a focused front view schematic of the locking device **900** in a locked configuration **902**.

(66) Cross-referencing FIGS. 7-10 and as most clearly seen in FIG. 9-10, the locking device **900** includes an axially actuatable shaft **910** coaxial with the pinion shaft **302** and located within the pinion shaft **302**. Specifically, the pinion shaft **302** is formed with an axial bore **309** that extends from the end **308** toward the differential **600**, and the axially actuatable shaft **910** is received within the bore **309**. Thus, the axially actuatable shaft **910** rotates with the pinion shaft **302**. Further, the locking device **900** includes a torque-to-thrust rotary mechanism **920** configured to move the axially actuatable shaft **910** between the unlocked configuration **901** and the locked configuration **902**. In an exemplary embodiment, the torque-to-thrust rotary mechanism **920** has an eccentric face **922** for contacting the axially actuatable shaft **910**. As shown, the contact between the torque-to-thrust rotary mechanism **920** and axially actuatable shaft **910** occurs at a rounded end **911** of the axially actuatable shaft **910** to provide a point contact and minimize rubbing friction. Also, the locking device **900** includes a rotary shift motor **930** configured to provide a high-torque low-speed output to rotate a shaft **935** coupled to the torque-to-thrust rotary mechanism **920**. The shaft **935** and torque-to-thrust rotary mechanism **920** rotate about an axis **931**.

(67) As shown, the locking device **900** may include a compressible member **940**, such as a spring, a ball **950**, a collar **960**, a biasing member **970**, such as a spring, and a pin **980**. The pin **980** is configured to connect the ball **950** to the collar **960** and to ride in a slot in the pinion shaft **302**. In FIG. 9, the axially actuatable shaft **910** is retracted. In FIG. 10, the rotary shift motor **930** has caused the torque-to-thrust rotary mechanism **920** to rotate 180 degrees to cause the axially actuatable shaft **910** to advance into the bore **309**. As result, the ball **950** is pushed by the spring **940** deeper into the bore **309**. As a result, the pin **980** is pushed in the same direction, causing the collar **960** to move into engagement with the projection **860** of the differential housing **650**. For example, teeth formed on the collar **960** may engage the projection **860**. As a result, the collar **960**

and shaft **302** are locked to the differential housing **650**.

(68) Further rotation of the torque-to-thrust rotary mechanism **920** causes the axially actuatable shaft **910** to be retracted from the bore **309**. Biasing element **970** may then force the disengagement of the collar **960** from the projection **860** of the differential housing **650** such that the locking device **900** returns to the unlocked configuration **901** of FIG. **9**. Thus, the differential **600** is open and may split torque along both paths **401** and **402**.

(69) Accordingly, a number of embodiments of electric drive units are provided with compact structures and compact arrangements. In exemplary embodiments, motor torque is split into two paths. Within each path, torque is again split between two transfer shaft assemblies that both are configured to power a same final drive gear. The meshes between each pinion gear and transfer shaft assembly pairs are arranged at 180 degree opposite angles which yields a net zero radial load on the pinion shafts. The net zero radial loads on the pinon shafts allows for smaller bearing for support of the shafts. The smaller bearings yield lower losses and higher efficiency. Further, the secondary stage transfer shaft assembly reductions to the final drive ring gears also carry twenty-five percent of the total torque each. This provides for smaller diameter transfer shaft assembly support bearings which further minimize losses and increase efficiency.

(70) In certain embodiments, the drive unit architecture described herein provides for up to 95% efficiency. In certain embodiments, the drive unit architecture described herein eliminates or reduces radial loads on motor pinion shafts. In certain embodiments, the drive unit architecture described herein minimizes loads on individual gears and reduces the required size of such gears. In certain embodiments, the drive unit architecture described herein provides for minimum bearing losses. In certain embodiments, the drive unit architecture described herein provides for a locking device to an upstream differential, i.e., between the motor and the gearing. In certain embodiments, the drive unit architecture described herein minimizes the mass of the differential. In certain embodiments, the drive unit architecture described herein may be utilized with chassis-mounted and beam-mounted applications.

(71) In summary, embodiments herein provide for minimum stress/maximum load sharing for the gears and differential while. Further embodiments provide for separating load cancellation, for minimum bearing loads, and for an upstream differential which is further provided with a locking device. In certain embodiments, an electric drive unit features a motor enclosed in a rotor with four transfer axles on two axes, four total axes, two final drive rings gears, and a locking differential.

(72) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. An electric drive unit comprising: an electric motor configured to provide an electric motor torque; an input differential configured to split the electric motor torque into a first path and a second path; a first final drive gear on the first path and a second final drive gear on the second path; a first pinion gear on the first path and a second pinion gear on the second path; a first rear transfer shaft assembly and a first front transfer shaft assembly on the first path; and a second rear transfer shaft assembly and a second front transfer shaft assembly on the second path; wherein the first pinion gear is configured to transfer torque to the first rear transfer shaft assembly and to the

first front transfer shaft assembly; wherein the first rear transfer shaft assembly and the first front transfer shaft assembly are both configured to transfer torque to the first final drive gear; wherein the second pinion gear is configured to transfer torque to the second rear transfer shaft assembly and the second front transfer shaft assembly; wherein the second rear transfer shaft assembly and the second front transfer shaft assembly are both configured to transfer torque to the second final drive gear; wherein the first pinion gear engages with the first rear transfer shaft assembly at a first rear interface and engages with the first front transfer shaft assembly at a first front interface; and wherein the first rear interface and the first front interface are separated by 180 degrees about the first pinion gear to yield a net zero bending load on the first pinion gear.

2. The electric drive unit of claim 1, wherein: each pinion gear rotates about a pinion axis; each rear transfer shaft assembly rotates about a rear axis; each front transfer shaft assembly rotates about a front axis; the pinion axis, the rear axis, and the front axis lie in a same plane; and the pinion axis is located between the rear axis and the front axis.

3. The electric drive unit of claim 1, wherein: each transfer shaft assembly comprises a transfer gear and a final drive pinion, wherein the transfer gear and the final drive pinion are mounted on a transfer shaft; each transfer gear meshes with a respective pinion gear; and each final drive pinion meshes with a respective final drive gear.

4. The electric drive unit of claim 3, wherein: the transfer shaft of the first rear transfer shaft assembly and the transfer shaft of the second rear transfer shaft assembly both rotate about a rear axis; and the transfer shaft of the first front transfer shaft assembly and the transfer shaft of the second front transfer shaft assembly both rotate about a front axis.

5. The electric drive unit of claim 1, wherein: the input differential is configured to transfer torque to a first pinion shaft and a second pinion shaft; the first pinion shaft and the second pinion shaft rotate about a first axis; the first pinion gear is mounted on the first pinion shaft; and the second pinion gear is mounted on the second pinion shaft.

6. The electric drive unit of claim 1, wherein: each pinion gear rotates about a pinion axis; each rear transfer shaft assembly rotates about a rear axis; each front transfer shaft assembly rotates about a front axis; the pinion axis, the rear axis, and the front axis define and lie in a single plane; and the pinion axis is located directly between and equidistant from the front axis and the rear axis.

7. The electric drive unit of claim 1, wherein: the first pinion gear is configured to split torque between the first rear transfer shaft assembly and the first front transfer shaft assembly such that the first rear transfer shaft assembly carries twenty-five percent of the electric motor torque and the first front transfer shaft assembly carries twenty-five percent of the electric motor torque; and the second pinion gear is configured to split torque between the second rear transfer shaft assembly and the second front transfer shaft assembly such that the second rear transfer shaft assembly carries twenty-five percent of the electric motor torque and the second front transfer shaft assembly carries twenty-five percent of the electric motor torque.

8. The electric drive unit of claim 1, further comprising a locking device configured to selectively connect and disconnect the input differential from the first pinion gear.

9. The electric drive unit of claim 8, wherein: the input differential is configured to transfer torque to a first pinion shaft; the locking device comprises an axially actuatable shaft coaxial with the first pinion shaft and located within the first pinion shaft.

10. The electric drive unit of claim 9, wherein the locking device comprises a torque-to-thrust rotary mechanism configured to move the axially actuatable shaft between an unlocked configuration and a locked configuration, and wherein the locking device comprises a rotary shift motor configured to rotate the torque-to-thrust rotary mechanism.

11. The electric drive unit of claim 8, wherein the first pinion shaft is formed with an axial bore; and the axially actuatable shaft is located within the axial bore of the pinion shaft.

12. The electric drive unit of claim 8, wherein: the input differential is configured to transfer torque to a first pinion shaft rotatable about a first axis; and the locking device comprises an axially

actuatable fork configured to move along a fork axis parallel to the first axis.

13. The electric drive unit of claim 1, wherein: the input differential is configured to transfer torque to a first pinion shaft and a second pinion shaft; each pinion shaft is axially constrained by an internal bearing press fit.

14. The electric drive unit of claim 1, further comprising a bearing arrangement of ten deep groove ball bearings and six cylindrical roller bearings.

15. An electric drive unit comprising: an electric motor configured to provide an electric motor torque; an input differential; a pinion gear, wherein the input differential is configured to selectively transmit torque to the pinion gear; a rear transfer shaft assembly and a front transfer shaft assembly, wherein the pinion gear is configured to transfer torque to the rear transfer shaft assembly and to the front transfer shaft assembly; a final drive gear, wherein the rear transfer shaft assembly and the front transfer shaft assembly are both configured to transfer torque to the same final drive gear; and a locking device configured to selectively connect and disconnect the input differential from the pinion gear, wherein: the input differential is configured to transfer torque to a pinion shaft; the pinion shaft is formed with an axial bore; and the locking device comprises an axially actuatable shaft coaxial with the pinion shaft and located within the axial bore of the pinion shaft.

16. The electric drive unit of claim 15, wherein: the pinion gear rotates about a pinion axis; the pinion gear meshes with the rear transfer shaft assembly at a rear interface; the pinion gear meshes with the front transfer shaft assembly at a front interface; and the pinion axis is located between the rear interface and the front interface.

17. A vehicle comprising: an electric motor configured to provide an electric motor torque; a first pinion gear on a first path and a second pinion gear on a second path; a first rear transfer shaft assembly on the first path and a second rear transfer shaft assembly on the second path; a first front transfer shaft assembly on the first path and a second front transfer shaft assembly on the second path; a first final drive gear on the first path and a second final drive gear on the second path; and a first wheel on the first path and a second wheel on the second path; wherein the first pinion gear is configured to transfer torque to the first rear transfer shaft assembly and the first front transfer shaft assembly; wherein the first rear transfer shaft assembly and the first front transfer shaft assembly are both configured to transfer torque to the first final drive gear, and the first final drive gear is configured to drive the first wheel; wherein the second pinion gear is configured to transfer torque to the second rear transfer shaft assembly and the second front transfer shaft assembly; wherein the second rear transfer shaft assembly and the second front transfer shaft assembly are both configured to transfer torque to the second final drive gear, and the second final drive gear is configured to drive the second wheel; and wherein torque from each pinion gear is split between the respective rear and front transfer shaft assemblies such that each transfer shaft assembly carries twenty-five percent of total motor torque.

18. The vehicle of claim 17, wherein: the first pinion gear rotates about a first pinion axis; the first pinion gear meshes with the first rear transfer shaft assembly at a first rear interface; the first pinion gear meshes with the first front transfer shaft assembly at a first front interface; the first pinion axis is located between the first rear interface and the first front interface.

19. The vehicle of claim 17, wherein: the first pinion gear engages with the first rear transfer shaft assembly at a first rear interface and engages with the first front transfer shaft assembly at a first front interface; the first rear interface and the first front interface are separated by 180 degrees about the first pinion gear to yield a net zero bending load on the first pinion gear.

20. The vehicle of claim 17, further comprising: an input differential configured to split the electric motor torque into the first path and the second path; a pinion shaft configured to receive torque from the input differential, wherein the pinion shaft is formed with an axial bore; and a locking device configured to selectively connect and disconnect the input differential from the pinion shaft,

wherein the locking device comprises an axially actuatable shaft coaxial with the pinion shaft and located within the axial bore of the pinion shaft.
